PROVINCIAL DIGITAL BLUEPRINT

Bataan Internet Infrastructure Plan

BatFIRST: Fiber Infra Resiliency & Sustainable Transformation

Operationalizing a Fiber PPP and DTIP Framework for Faster, Cheaper, Stable, and Highly Reliable Connectivity Across Bataan

Actionable Connectivity Mandates

Faster

Deploying low-loss underground microduct fiber with localized edge caching to keep provincial traffic within sub-10ms response times.

Cheaper

Absorbing civil trenching via government public works, reducing entry barriers for local ISPs and driving down household tariffs.

More Stable

Direct 48V DC solar-battery bus connection isolates telecom nodes from grid voltage surges, avoiding power inverter failures.

Reliable

Closed-loop underground municipal fiber rings ensure automatic sub-second rerouting if a physical cable cut occurs.

The DTIP Regulatory Catalyst

RA 12234 Regime Shift

The Konektadong Pinoy Act classifies digital infrastructure operators as Data Transmission Industry Participants (DTIP), eliminating the 25-year Congressional franchise bottleneck.

Market entry shifts to streamlined administrative NTC registration (up to 10 years for middle mile, 5 years for last mile), drastically lowering regulatory risk and capital deployment lag.

Mandatory Open Access

RA 12234 legally prohibits anti-competitive hoarding of passive infrastructure on the national Access List, requiring access on fair, reasonable, and non-discriminatory (FRAND) terms.

Enforces statutory coordination across utility agencies via a national "Dig-Once" doctrine, creating legal backing for unified multi-duct underground conduits.

The Tollway Concession Analogy

Government Builds, OpCo Manages

Like public expressways (e.g., SCTEX), the government finances civil trenching, manholes, and multi-duct pipes as permanent public capital assets.

The private DTIP concessionaire deploys active electronics, operates the NOC, maintains 99.9% SLAs, and leases capacity to downstream retail providers.



Strategic Linear Rights-of-Way

Transmission

NGCP ground easements and railway corridors provide clear, continuous linear paths free from municipal traffic interference and road widening.

***transmission line property is owned by TRANSCO. Operations was given to NGCP not sure if telco operations was also given to them.*

River Retaining Walls

DPWH and provincial flood-control retaining walls offer stable, concrete-reinforced conduits along natural linear waterways through town centers.

***most of development are beside the river even the earlier times. What if we include dark fiber in the plans.*

Irrigation Corridors

NIA service roads and canal dikes form unpaved, dig-ready pathways linking agricultural barangays directly to major municipal centers.

***it runs through agricultural properties across various municipalities.*

Bypasses & Sidewalks

New highway bypass alignments and urban sidewalk utility strips enable shallow trenching ahead of commercial real estate development.

***Under construction road with plans already in place, minimizes the cost of digging.*

****What if we have a map for; 1. Strategic Linear ROW 2. Existing Telco Backbone Lines. Determine Existing, Weak, or Missing Points. Considering Demand and Future Plans**

ROW Challenges: If government or private will be covered will they enter into sale or usufruct with province or LGU? LandCo and OpCo setup?

Standardized Civil Duct Profiles

Road Openings & Sidewalks

Primary Infrastructure: 4-Way 14/10mm HDPE microduct bundle encased in sand bedding and lean concrete at 0.8m–1.2m depth with magnetic warning tape.

Sidewalk Micro-Trenching: 2-Way 12/9mm microducts cut along curb lines (300mm deep, elastomeric sealant) for low-impact, rapid deployment in paved town centers.

Composite Handholes: Pre-cast polymer handholes placed every 300m for rapid fiber blowing and mid-span access without pavement breaking.

Irrigation, Rail & Retaining Walls

River Retaining Wall Brackets: Galvanized steel or heavy-duty UPVC conduit runs anchored directly to concrete wall coping above historical high-water marks.

Irrigation & Rail Corridors: Direct-buried armored HDPE conduits laid at 1.0m depth along canal service dikes and transmission easement perimeter lines.

Expansion Capacity: Continuous reservation of at least 2 spare sub-ducts for municipal telemetry, flood sensors, and carrier-of-carriers leasing.

****Detailed Engineering Design that fits the ROW condition. From this we can come up with a costing**

Policy & Environmental Framework

- **Mandatory Dig-Once Ordinance:** Requires DPWH, provincial engineering, and utility developers to lay multi-duct banks during any road construction, reducing excavation capex by up to 80%.
- **Anti-Spaghetti Wire Mandate:** Regulates aerial clutter to prevent pole overloading, mitigate fire hazards, and streamline emergency maintenance for electric cooperatives (PENELCO).
- **Mandatory Cable Tagging & Asset Audits:** Operators must register RFID tags on overhead spans; untagged, dead, or disconnected cables are declared public hazards subject to summary removal.
- **DENR E-Waste Compliance (DAO 2013-22):** Abandoned telecom cables are classified as hazardous WEEE (Waste Electrical & Electronic Equipment), mandating safe recovery through DENR-accredited TSD facilities.

****Just some policy push to create demand and relevance for the planned fiber line.**

The Last-Mile Open Ecosystem

The Wheeling Model

Like an electrical grid charging a wheeling fee for power transmission, the middle-mile concessionaire charges a transparent, volume-based transport tariff per Gbps-kilometer.

Local micro-ISPs connect into standardized local aggregation Points of Presence (POPs), bypassing the prohibitive expense of deploying long-haul backbones.

Hyper-Local Retailers

Community micro-ISPs, cable TV operators, and wireless providers focus on retail execution: sachet Wi-Fi tokens, ₱500 micro-broadband, and local customer service.

Open access transforms local ISPs from fragmented operators into long-term commercial tenants, driving rapid retail broadband penetration in underserved barangays.

****Dark or Lit Fiber? What business model would work for the O&M?**

Concession vs Incumbent Telco

Strategic Dimension	Traditional Telco Model	BatFIRST Concession Model
Civil Capital Source	100% Private CAPEX on corporate balance sheet	Public civil works & RROW integration (PPP/LGU)
Infrastructure Access	Proprietary, closed, duplicated builds	Mandated open-access multi-tenant leasing
Typhoon Survivability	Vulnerable overhead aerial drops on poles	Underground microducts & off-grid solar nodes
Barangay Retail Tariffs	Rigid post-paid packages (₱1,200+/mo)	Sachet digital vouchers & micro-broadband
Disaster Redundancy	Single highway route; frequent backhoe cuts	Closed underground rings + secondary wireless links

Bankability & Infra Financing

65%

Contracted Base Off-Take

De-risking for Multilaterals & Private Capital

Anchored Revenue Base: Locks in 5–10 year off-take commitments across provincial e-Gov, LGU DRRM Calamity Funds, and enterprise dark fiber IRUs, securing predictable debt service coverage.

Asset-Light Balance Sheet: Public co-financing of long-life passive civil conduits insulates the private OpCo from heavy civil debt, delivering high DSCR (>1.4x) demanded by ADB, IFC, and commercial lenders.

Commercial Upside: Blends contracted availability fees with scalable wheeling tariffs from last-mile ISPs and tower-sharing leases at Resiliency Center nodes.

****Just thinking on how to lock in revenue contracts to be bankable.**
Like grid off-takes for solar (I don't know if its comparable)

MOVING FORWARD

Next Steps & Discussion

Operationalizing the BatFIRST Open-Access Concession for Resilient Provincial Connectivity